

Módulo 6 - Evidências: do primeiro contêiner ao rollback em Kubernetes

1. Versões das Ferramentas

Comandos executados: `docker --version`, `kind --version` e `kubect1 version --client`

```
Client:
  Version:      28.0.4
  API version:  1.48
  Go version:   go1.23.7
  Git commit:   b8034c0
  Built:        Tue Mar 25 15:07:48 2025
  OS/Arch:      windows/amd64
  Context:      desktop-linux

Server: Docker Desktop 4.40.0 (187762)
Engine:
  Version:      28.0.4
  API version:  1.48 (minimum version 1.24)
  Go version:   go1.23.7
  Git commit:   6430e49
  Built:        Tue Mar 25 15:07:22 2025
  OS/Arch:      linux/amd64
  Experimental: false
containerd:
  Version:      1.7.26
  GitCommit:    753481ec61c7c8955a23d6ff7bc8e4daed455734
runc:
  Version:      1.2.5
  GitCommit:    v1.2.5-0-g59923ef
docker-init:
  Version:      0.19.0
  GitCommit:    de40ad0

kind v0.33.0 go1.26.7 windows/amd64

Client Version: v1.32.2
Kustomize Version: v5.5.0
```

2. DockerFile e deployment.yaml

```
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt ./
RUN python -m pip install --no-cache-dir -r requirements.txt
COPY app.py ./
RUN useradd --create-home --uid 10001 app
USER app
EXPOSE 8000
```

```
CMD ["python", "-m", "uvicorn", "app:app", "--host", "0.0.0.0", "--port", "8000"]

apiVersion: apps/v1
kind: Deployment
metadata:
  name: hospital-api
spec:
  replicas: 3
  selector:
    matchLabels:
      app: hospital-api
  template:
    metadata:
      labels:
        app: hospital-api
    spec:
      containers:
      - name: api
        image: hospital-api:1.0.0
        imagePullPolicy: Never
        ports:
        - containerPort: 8000
---
apiVersion: v1
kind: Service
metadata:
  name: hospital-api-service
spec:
  type: NodePort
  selector:
    app: hospital-api
  ports:
  - port: 8000
    targetPort: 8000
    nodePort: 30000
```

3. Saída da validação com --dry-run=client

Comandos executados: `kubectl apply -f deployment.yaml --dry-run=client`

```
deployment.apps/hospital-api created (dry run)
service/hospital-api-service created (dry run)
```

4. Contexto kind-hospital-local confirmado.

Comandos executados: `kubectl config current-context`

```
kind-hospital-local
```

5. Imagem carregada no cluster.

Comandos executados: `kubectl config current-context`

```
kind-hospital-local
```

6. Rollout inicial concluído.

Comandos executados: `kubectl apply -f deployment.yaml, kubectl rollout status deployment/hospital-ap`

```
deployment.apps/hospital-api created
service/hospital-api-service created
Waiting for deployment "hospital-api" rollout to finish: 0 of 3 updated replicas
are available...
Waiting for deployment "hospital-api" rollout to finish: 1 of 3 updated replicas
are available...
Waiting for deployment "hospital-api" rollout to finish: 2 of 3 updated replicas
are available...
deployment "hospital-api" successfully rolled out
```

7. Lista de Pods e do Service.

Comandos executados: `kubectl get pods,svc`

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	26m

9. listagem de Pods antes e depois de apagar um à mão.

Comandos executados: `kubectl get pods -n hospital, kubectl get pods -n hospital, kubectl get pods -n hospital`

NAME	READY	STATUS	RESTARTS	AGE
hospital-api-86455444c6-nzdqc	1/1	Running	0	29m
hospital-api-86455444c6-tmdpg	1/1	Running	0	29m

pod "hospital-api-86455444c6-nzdqc" deleted

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

hospital-api-86455444c6-tmdpg	1/1	Running	0	29m
hospital-api-86455444c6-tp7lx	0/1	Running	0	1s

10. Trecho do describe com ImagePullBackOff.

Comandos executados: `kubectl describe pod hospital-api-86455444c6-tmdpg -n hospital`

```
Name:                hospital-api-86455444c6-tmdpg
Namespace:           hospital
Priority:             0
Service Account:     default
Node:                hospital-local-control-plane/172.18.0.2
Start Time:          Sun, 20 Sep 2026 11:42:45 -0300
Labels:              app=hospital-api
                    pod-template-hash=86455444c6
Annotations:         <none>
Status:              Running
IP:                  10.244.0.6
IPs:
  IP:                10.244.0.6
Controlled By:       ReplicaSet/hospital-api-86455444c6
Containers:
  hospital-api:
    Container ID:     containerd://eed6983028a7daaca5a82a68a57be2116003f903b1226f2a2f28212309bd6e95
    Image:             hospital-api:1.0.0
    Image ID:          docker.io/library/import-2026-09-20@sha256:e33a4a4661bef6732cb0b612fd16859d9cb8bdaa2b798244881ec1adcf372b35
    Port:              8000/TCP
    Host Port:         0/TCP
    State:              Running
      Started:          Sun, 20 Sep 2026 11:42:46 -0300
    Ready:              True
    Restart Count:      0
    Limits:
      cpu:               250m
      memory:            256Mi
    Requests:
      cpu:               100m
      memory:            128Mi
    Liveness:            http-get http://:http/health/live delay=5s timeout=1s period=5s
                        #success=1 #failure=3
    Readiness:           http-get http://:http/health/ready delay=2s timeout=1s period=3s
                        #success=1 #failure=3
    Environment Variables from:
      hospital-api-config ConfigMap Optional: false
    Environment:         <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-xzbqj
(ro)
Conditions:
```

```

Type                                Status
PodReadyToStartContainers          True
Initialized                         True
Ready                              True
ContainersReady                    True
PodScheduled                       True
Volumes:
  kube-api-access-xzbqj:
    Type:                                Projected (a volume that contains injected data from
multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:                kube-root-ca.crt
    ConfigMapOptional:            <nil>
    DownwardAPI:                  true
QoS Class:                           Burstable
Node-Selectors:                      <none>
Tolerations:                        node.kubernetes.io/not-ready:NoExecute op=Exists for
300s
                                   node.kubernetes.io/unreachable:NoExecute op=Exists
for 300s
Events:
  Type    Reason      Age   From          Message
  ----    -
Normal    Scheduled   40m   default-scheduler Successfully assigned
hospital/hospital-api-86455444c6-tmdpg to hospital-local-control-plane
Normal    Pulled      40m   kubelet       Container image "hospital-api:1.0.0"
already present on machine and can be accessed by the pod
Normal    Created     40m   kubelet       Container created
Normal    Started     40m   kubelet       Container started

```

11. Comando e status do rollback.

Comandos executados: `kubectl rollout status deployment/hospital-api -n hospital`

```
deployment "hospital-api" successfully rolled out
```

12. Resposta de /health/live após o rollback.

Comandos executados: `curl.exe --fail --silent http://127.0.0.1:18080/health/live`

```
{"status":"live"}
```

13. Confirmação da remoção do cluster.

Comandos executados: `kind delete cluster --name hospital-local`

```
Deleting cluster "hospital-local" ...  
Deleted nodes: ["hospital-local-control-plane"]
```